

# Glossary

- Abrasion/corrosion** The wearing away of the river bed and banks by the river's load hitting them repeatedly.
- Accessibility** The ease with which a place can be reached. An area with high accessibility will generally have a well-developed transport network and be centrally located.
- Active volcano** A volcano that is currently erupting or has recently erupted and is likely to erupt again.
- Adaptation (to climate change)** Methods of learning to live with climate change, e.g. sea walls and malaria tablets.
- Adult literacy rate** The percentage of an adult population with basic reading and writing skills.
- Aeroponics** A system where plants are never placed into water, with crops growing suspended in the air. Nutrients are supplied from a mist that is sprayed on to their roots. The entire process is enclosed, so the nutrient mist is able to remain around the plants for longer.
- Agglomeration economy** The advantages that firms gain when they are located near other similar and related firms, such as cost savings from shared infrastructure.
- Agricultural productivity** The ratio of agricultural inputs to outputs.
- Agricultural region** An area with relatively uniform physical characteristics (altitude, climate, soil type), where because of these characteristics a particular type of farming dominates, such as wheat, corn or cotton.
- Agricultural technology** The application of techniques to control the growth and harvesting of animal and vegetable products.
- Agriculture** The production of crops and livestock, aquaculture and forest products, e.g. rice, fish and cotton, for food and non-food products.
- Agro-industrialisation** Industrialised farming that is typically large scale and capital intensive.
- Air pressure** The pressure at any point on the Earth's surface that is due to the weight of the air above it; it decreases as altitude increases. At sea level the average pressure is 1013 millibars (mb). Areas of relatively high pressure are called anticyclones; areas of low pressure are called depressions.
- Albedo** The reflectivity of a surface – snow and ice are very reflective, whereas dark surfaces, such as soil, are not.
- Alluvium** Material deposited by rivers in a channel or on a floodplain.
- Altitude** Measurement of height, usually given in metres above sea level. Temperature declines, on average, 1°C for every 100 m (and therefore rises 1°C with every decrease in altitude of 100 m).
- Annotation** Descriptive or explanatory labelling of a photo or diagram to show the main characteristics/features.
- Anti-natalist policy** A policy that is discouraging and unsupportive of increasing fertility.
- Aquaponics** A system that combines fish farming (aquaculture) with hydroponics. Fish waste is naturally converted into nutrients for plants, while the plants filter the water for the fish. This creates a closed-loop ecosystem.
- Aquifer** A rock that allows water to move through it, such as a layer of sandstone.
- Arable farming** The cultivation of crops.
- Arch** A natural bridge-like feature formed by erosion. Arches are formed from the erosion of a headland where two caves meet and break through the headland.
- Archipelagic country** A country made up of a group of islands, interconnecting waters, and other natural features that are closely interrelated.
- Arid area** Usually defined as an area that receives less than 250 mm of rainfall each year.
- Ash fall** Very small, hard, jagged material of volcanic origin. It is the most frequent and widespread volcanic hazard.
- Atlas maps** A collection of maps usually showing different countries, continents, and physical and human geography.
- Attrition** The process by which particles of rock being transported by a river are rounded and become smaller in size by being struck against one another. Particles near the shoreline become smaller and more rounded due to more frequent attrition.
- Backwash** The movement of water back down the beach due to the effect of gravity.
- Bar** A depositional feature – a long ridge of sand or pebbles running parallel to a coastline that is submerged at high tide. Some bars develop as offshore bars when waves disturb sediments on the sea bed and form them into a submarine ridge or bar, while others form from the development of a spit across the whole of a small bay.
- Base map** A very simple map, often just an outline, which can then be filled in to show some characteristics of the place in question.
- Battery energy storage system (BESS)** A battery system that enables excess energy from renewables to be stored and then released when the power is needed most.
- Bay** A wide, open, curving indentation of the sea.
- Beach** A feature of coastal deposition, consisting of pebbles on exposed coasts or sand on sheltered coasts. It is usually defined by the high and low water marks.
- Bedload** Material rolled along the river bed.
- Big data** Extremely large data sets analysed computationally to reveal patterns, trends and associations.
- Bimodal** A distribution of a population that has two modes (most frequent values).
- Biodiversity** Biological diversity is a measure of the variety of the Earth's plant and animal species, of genetic differences within species, and of the ecosystems that support those species.
- Biofuel** Any fuel produced from organic matter, either directly from plants or indirectly from industrial, commercial, domestic or agricultural wastes. For example fossil fuel substitute can be made from a range of crops, including oilseeds, wheat and sugar.
- Birth rate** The number of live births per thousand population in a year.
- Body wave** A seismic wave that travels through the Earth's interior.
- Bradshaw model** A geographical model that suggests how a river's characteristics change from the source to the mouth of the river.
- Braiding/braided channel** The sub-division of a river into several channels, caused by deposition of sediment as small islands in the main channel.



**Break-of-bulk point** A location, such as a sea port, where freight has to be transferred from one mode of transport to another.

**Business/retail park** Similar to an industrial estate, but dominated by tertiary activities.

**Capacity (rivers)** The total amount of sediment a river can carry.

**Capital intensive** A high level of investment in plant and machinery compared to labour.

**Capitalism** The social and economic system that relies on the market mechanism to distribute the factors of production (land, labour, capital) in the most efficient way.

**Carrying capacity (of a populated area)** The largest population that the resources of a given environment can support.

**Carrying capacity (of a tourism destination)** The maximum number of people that may visit a tourism destination at the same time without causing destruction of the physical, economic or sociocultural environment and an unacceptable decrease in the quality of visitor satisfaction.

**Cartoon** A diagram that may exaggerate a feature to draw attention to a particular point.

**Catchment management** The regulations and policies that govern the oversight of surface and groundwater resources within the area of a river catchment.

**Cave** Coastal caves are large holes formed where relatively soft rock containing lines of weakness is exposed to severe wave action.

**Channel flow** The movement of water within the river channel.

**Channel network (river system)** The pattern of a main river and its tributaries within a drainage basin.

**Channel roughness** The frictional force of a river bed. A rough bed (with boulders, pebbles, potholes) exerts more friction than a smooth channel.

**Child mortality rate** The number of deaths of children under five years of age per thousand live births per year.

**Cinder cone volcano** A steep-sided volcano made of fragments of volcanic materials, e.g. cinders and tephra.

**Circular economy** An economic system based on the reuse and regeneration of materials and products, especially as a means of continuing production in a sustainable way.

**Clean air transition** The period of change from heavy reliance on fossil fuels to most of people's energy needs being supplied by renewable sources of energy.

**Cliff** A rock-face along a coastline where coastal erosion, weathering and mass movements are active and the slope rises steeply (over 45°) and for some distance. The nature of the cliff depends on the nature of the rocks, their hardness and their jointing pattern.

**Climate** The combination of weather conditions at a particular place over a period of time – usually a minimum of 30 years. Climate thus includes the averages, extremes and frequencies of all meteorological elements, such as temperature, atmospheric pressure, precipitation, wind, humidity and sunshine.

**Climate loss** Pre-harvest food losses from: not planting a crop, or at least not the desired crop; changing crops due to weather; extreme weather reducing or wiping out a crop before harvest; increased predation, pests and/or diseases resulting from climate change.

**Closed system** A system unconnected to other entities. It has no inputs from, or outputs to, elsewhere.

**Cloud** Water vapour condensed into minute water particles that float in the atmosphere. Clouds are formed by the cooling of air containing water vapour, which generally condenses around tiny dust or ice particles.

**Coastal erosion** The wearing away of coastlines by moving agents, such as waves and wind, to create coastal features such as cliffs and wave-cut platforms.

**Coastal management strategy** A measure taken to prevent coastal erosion and/or flooding. To reduce erosion, several different forms of coastal protection are used. These can be divided into hard engineering and soft engineering.

**Coastline** The area of contact between land and sea.

**Colonialism** The establishment of colonies in one or a number of territories by people from another territory.

**Commercial farming** Farming for profit, where food is produced for sale in the market.

**Commodity chain** The stages involved in manufacturing a finished product (commodity) for sale to consumers. These stages may occur in factories in different countries.

**Community energy** Energy produced close to the point of consumption.

**Community tourism** A form of tourism that aims to include and benefit local communities, particularly in developing countries.

**Competence (rivers)** The maximum size of particle a river can carry.

**Composite cone volcano** A steep-sided volcano formed of alternating layers of lava and volcanic fragments, e.g. cinders and tephra.

**Concordant coastline** A relatively straight coastline in which the relief and geology are parallel to the coast.

**Condensation** The process by which water vapour changes into water droplets.

**Confluence** The point where a tributary stream/river joins a main river.

**Conservation** Allowing for developments that do not damage the character of a location.

**Conservation of living resources** The management of the human use of natural resources to provide the maximum benefit to current generations while maintaining capacity to meet the needs of future generations.

**Conservative/transform boundary** A plate boundary in which plates move past each other in different directions or at different speeds but no material is destroyed or created. They are associated with earthquake activity.

**Constructive wave** A wave with a long wave length and a low height. Constructive waves help to build up beaches by deposition.

**Consumer culture** A society in which patterns of consumption are a key basis for personal identity and status differentiation.

**Containerisation** A system of intermodal freight transport using standard international containers.

**Contour ploughing** Prevents or diminishes the downslope movement of water and soil by ensuring that ridges and furrows are at right angles to the slope.

**Convictional rainfall** Rainfall associated with hot climates, resulting from the rising of convection currents of warm air. Air that has been warmed by the extreme heating of the ground surface rises to great heights and is cooled quickly. The water vapour carried by the air condenses and rain falls heavily. Convictional rainfall is usually associated with thunderstorms.



**Convergent/collision zone** A plate boundary where two plates meet. It may lead to the creation of fold mountains, e.g. the Himalayas.

**Convergent/destructive boundary** A plate boundary where two plates meet and the denser oceanic plate subducts/plunges underneath the less dense continental plate. They are associated with creating ocean trenches and fold mountains.

**Coral** Living organisms that may form large reefs. Coral reefs provide a habitat for a wide diversity of living organisms.

**Corruption** Dishonest or illegal behaviour by people in powerful positions, typically involving bribery.

**Counterurbanisation** The movement of people away from larger urban areas to smaller urban areas and/or rural settlements.

**Country of destination** The country where a migration is completed.

**Country of origin** The country from which a migration begins.

**Crater** A bowl-shaped depression at the summit of a volcano, formed by a volcanic eruption.

**Cultural diffusion** The spreading of cultural traits from one place to another.

**Cultural identity** The definition of a group or individual (by themselves or others) in terms of ethnicity, nationality, language, religion and gender.

**Culture** The inherited ideas, beliefs, values and knowledge of a society.

**Death rate** The number of deaths per thousand population in a year.

**Debt** Money owed by a country to another country, to private creditors (e.g. commercial banks), or to international agencies such as the World Bank or IMF.

**Debt relief** The cancellation of debts owed by developing nations to developed countries or to institutions such as the World Bank, in order to allow the government to shift funds toward social development.

**Debt service ratio** The proportion of a country's GDP required to pay off its debts each year.

**Decentralisation** The outward movement of economic activity from CBDs and inner cities.

**Decimal** A way of expressing 'inbetween' numbers. Decimal comes from the Latin word *decimus*, meaning *tenth*. Thus, decimals refer to tenths between whole numbers.

**Deep-focus earthquake** An earthquake that occurs at a depth of over 70 km beneath the Earth's surface.

**Deep-sea port** A port that has the depth of water, natural or dredged, and the technology to handle very large ocean-going ships, particularly ultra-large container ships.

**Deforestation** The removal of forest cover or destruction of forest for timber, fuel or charcoal burning, or clearing for agriculture and extractive industries such as mining. It can be complete or partial. It causes fertile soil to be blown away or washed into rivers, leading to soil erosion, drought, flooding and loss of wildlife.

**Deindustrialisation** The long-term absolute decline of employment in manufacturing.

**Delta** A flat, low-lying deposit of sediment that is found at a river's mouth.

**Demographer** A person who studies human populations.

**Demographic dividend** The growth in an economy resulting from a change in the age structure of a country's population.

**Demographic preparedness** The idea that there is time to enact policies and encourage behaviours to limit the potential adverse effects of demographic changes.

**Demographic transition model** A model illustrating the historical shift of birth and death rates from high to low levels in a population.

**Dependency ratio** The ratio of the number of people aged under 15 years and over 64 years to those in the 15–64 age group.

**Dependency theory** A theory that blames the relative underdevelopment of the developing world on exploitation by the developed world, first through colonialism and then by neo-colonialism.

**Deposition** The laying down of material carried by rivers or the sea because of a reduction of velocity or discharge (both causing a loss of energy), often caused by increased friction with vegetation or coarse particles.

**Detritivores** Organisms that feed on dead organic matter (detritus).

**Desert** A dry area with limited vegetation. Deserts can be either hot or cold. Characteristics common to all deserts include irregular rainfall of less than 250 mm per year.

**Desertification** The transformation of land that could once support human life into desert.

**Desire lines** A type of flow line that shows direction of travel.

**Destination footprint** The environmental impact caused by an individual tourist on holiday in a particular destination.

**Destructive wave** A wave with a high height and a short wave length, which helps erode beach materials and cliffs.

**Development** The use of resources to improve the quality of life in a country.

**Development gap** The difference in income and quality of life in general between the highest and lowest-income countries in the world.

**Diaspora** The dispersion of a people, originally belonging to one country, around the world.

**Diet** The kinds of foods that a person, animal or community habitually eats.

**Digital divide** The gulf between those who have ready access to computers and the internet and those who do not.

**Direct channel precipitation** Rainfall and snow falling directly into a river or stream.

**Discharge** The amount of water passing a specific point at a given time (the volume times the velocity). It is measured in cubic metres per second.

**Discordant coastline** A coastline in which the relief and geology are at right angles to the coastline.

**Disruptive technology** An innovation that significantly alters the way that consumers, industries and businesses operate.

**Distribution maps** Maps that show the distribution of a geographic element in a given area. They can range from small scale, e.g. the distribution of types of shops along a street, to very large scale, e.g. the distribution of global biomes/ecosystems.

**Divergent/constructive boundary** A plate boundary where new material is created and two plates move away from each other, e.g. the North American and European Plates moving away from the Mid-Atlantic Ridge.

**Do minimum** A form of managing natural processes that involves very limited human intervention but may aim at protecting human lives.

**Do nothing** A form of managing natural processes that involves taking no action but allowing natural processes to operate without any human intervention.